DIC06
Effective Gate Capacitance

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 6 (PDF pages 24–25) to compare real MOS input loading with the handout's 8/16/24/32 fF fixed-capacitance approximations.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: four Table 6.1 cases (24–25)	01_gate_load: NL=1–4	All four cases simulated
Compare MOS and fixed-C waveform/delay (25)	8 fF per doubled MLOAD	Differences quantified; equality not assumed

3 Circuit and design derivation

Both drivers use $W_n = 0.6\,\mu$m, $W_p = 1.8\,\mu$m and $L = 0.25\,\mu$m. Each MOS load doubles the widths to $1.2/3.6\,\mu$m. NL identical loads connect in parallel to the same driver output; MOS multiplicity represents these parallel units while their output still switches. The reference case a used a unit load, and case b cascaded loads. Both are corrected to the doubled parallel loading specified in the handout.

The estimate is $C_g \approx C_{ox}(W_n + W_p)L$, with $C_{ox} = \epsilon_{ox}/t_{ox}$. The handout's 5.4 nm oxide gives about 7.87 fF per load only for its original process. The supplied encrypted model does not expose a verified oxide thickness. The handout's prose reverses W/L labels; W denotes width and L length here.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not accep-

tance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Input edges are 0.1 ns with a 5 ns period; transient duration is 10 ns with 1 ps printing. Both output 10–90% transitions and input/output 50% propagation delays are measured.

5 Results and specification checks

All four cases solved and measured successfully. Delays below are in ps; MOS/C denote real loads/fixed capacitors.

NL	C (fF)	$t_{pHL,MOS}$	$t_{pHL,C}$	$t_{pLH,MOS}$	$t_{pLH,C}$
1	8	76.80	63.82	139.95	88.76
2	16	112.89	85.88	241.42	127.05
3	24	148.64	107.34	339.83	164.48
4	32	184.10	128.38	439.90	201.64

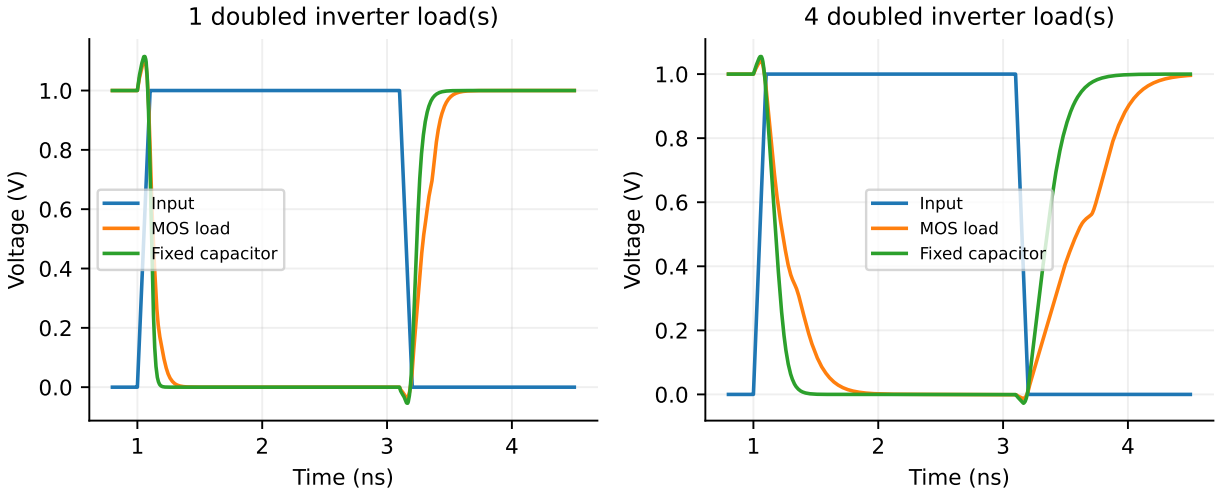


Figure 1: Waveforms with one and four loads; redrawn from this run's HSPICE data.

6 Discussion

The real MOS load is slower than the handout's fixed capacitor in all four cases, particularly for charging: NL=4 gives 439.90 ps versus 201.64 ps with 32 fF. Gate capacitance, overlap/Miller coupling and movement of the load output contribute; a single $C_{ox}WL$ value cannot capture every state. The handout's approximate equality is not reproduced under this model, a documented process-adaptation difference rather than an omitted experiment.

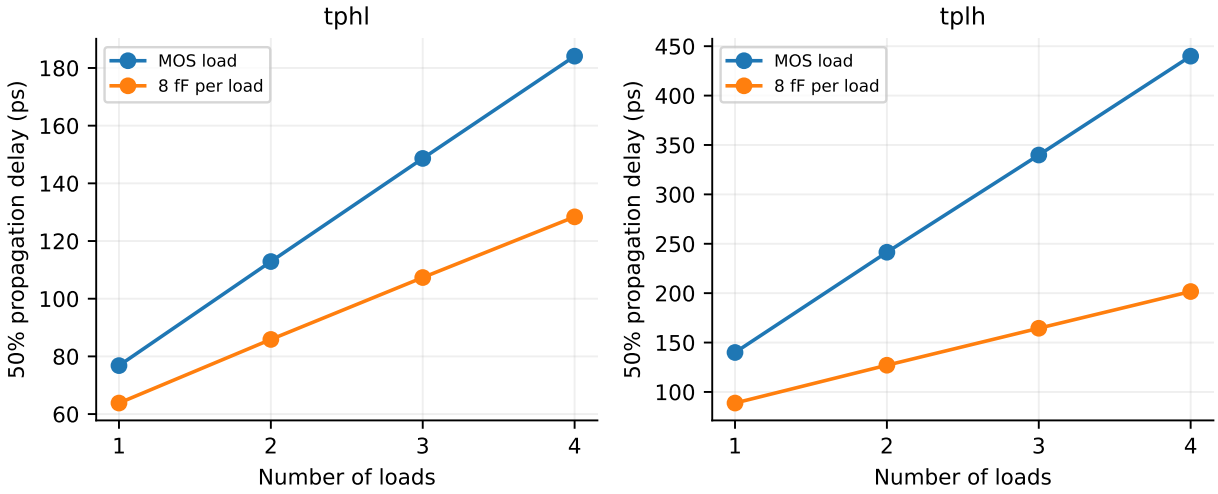


Figure 2: Propagation delay for the two load models; redrawn from this run's HSPICE data.

7 Conclusion

All four requested comparisons are complete. Eight fF per load is not a validated equivalent for this model. A subsequent cell-library model would require fitting across input slew, output loading and logic state.